

Umar Al Mukhtar Ibrahimkutty

Full Stack Developer | Backend Engineer | AI Systems Developer
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PROFILE SUMMARY

Computer Science undergraduate with experience building production-scale backend systems, AI-integrated applications, and full-stack web platforms. Developed software deployed for Smart City operations, engineered backend infrastructure serving thousands of users, and built an award-winning desktop application for automated video editing. Strong foundation in web technologies, distributed systems concepts, APIs, databases, and software engineering practices.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, C	Tools & Platforms: Git, GitHub, Vercel, FFmpeg, Tauri, Google Apps Script
Frontend: React.js, Next.js, Tailwind CSS, Three.js	Databases: MongoDB, PostgreSQL, MySQL, Prisma, Supabase
Backend: Node.js, Express.js, REST APIs, JWT Authentication	ML / CV: YOLOv11, OpenCV, Groq Vision API, LLM Integration

EXPERIENCE

Backend Developer	Smart City Project – TVM ICCC	Dec 2024 – Present
- Developed scalable REST APIs using Node.js, Express.js, and MongoDB for a Smart City waste detection platform deployed at the Thiruvananthapuram Integrated Command and Control Centre. - Integrated YOLOv11 and OpenCV computer vision pipelines with backend services through asynchronous inference workflows. - Designed structured logging, monitoring, and error-handling systems to improve production reliability. - Collaborated with frontend teams to deliver real-time analytics dashboards for city administrators.		
Backend Intern	Mulearn Foundation	Jan 2026 – Present
- Developed and maintained backend API endpoints supporting core platform infrastructure. - Refactored existing services and contributed to production deployments with the engineering team. - Participated in code reviews, debugging, and backend architecture discussions.		
Web Developer	Kasaroam Tourism Platform	Jan 2025 – Present
Built component-based frontend features for a Smart City-backed tourism platform, optimising responsiveness and page load performance for production deployment.		

PROJECTS

FocusFlow — Top 5 Winner, Codex Community Hackathon 2026	Rust • Tauri • React • FFmpeg
- Built a desktop application that transforms raw screen recordings into professional demo videos through automated click-based zooming and camera movements. - Engineered recording pipelines using Rust and FFmpeg with automatic timeline generation and post-processing workflows. - Implemented click tracking, automated editing logic, and export systems for streamlined video production. - Selected among the Top 5 winning projects in the Codex Community Hackathon 2026.	

MatchUndo Next.js • PostgreSQL • Prisma • Supabase

- Designed and launched a community platform connecting football fans for shared match-viewing events.
- Implemented authentication, moderation workflows, analytics tracking, SEO optimization, and venue discovery features.
- Built and deployed the complete product independently from architecture through production release.

Smart City Waste Detection Dashboard Node.js • Express.js • MongoDB • YOLOv11 • OpenCV

- Engineered backend architecture and AI integration for a city-scale litter detection system deployed in production.
- Designed APIs, inference pipelines, and data-processing services supporting real-time monitoring.

MediScribe – Handwritten Prescription Digitiser Python • Flask • Groq Vision API

- Built an AI-powered prescription digitization platform converting handwritten prescriptions into structured medical data.
- Developed image-processing, prompt-engineering, validation, and confidence-scoring pipelines

Inventory Manager Next.js • PostgreSQL • Prisma

- Developed inventory management software supporting purchases, consumption tracking, and asset management.
- Designed full-stack workflows currently used in production by a local business.

LEADERSHIP & ACTIVITIES

Chief Technology Officer — IEDC LBS College of Engineering

Mar 2024 – Dec 2025

Technology & Platform Head — IEDC Summit 2025

Sep 2025 – Dec 2025

Chair — IEEE Sensors Council SBC LBSCEK

Nov 2025 – Present

Webmaster — IEEE IA/IE/PELS Joint Chapter Kerala Section

Feb 2026 – Present

EDUCATION

B.Tech – Computer Science and Engineering

LBS College of Engineering, Kasaragod, Kerala

Expected Graduation: June 2027

CERTIFICATIONS

- Frontend Developer (React) — HackerRank
- Python Basics — HackerRank
- JavaScript Algorithms and Data Structures — freeCodeCamp
- Front-End Development Libraries — freeCodeCamp
- Responsive Web Design — freeCodeCamp
- Prompt Engineering Certification

ACHIEVEMENTS

- Top 5 Winner — Codex Community Hackathon 2026
- Led technology infrastructure supporting 5,000+ event participants.
- Built and deployed production systems used by Smart City and organizational stakeholders.

INTERESTS

Open Source • Distributed Systems • AI Applications • Software Architecture • Competitive Programming